

Lesson 24 — Make Your Own Loops

Matches LEGO Teacher Portal lesson: "Flowers for Croc".

Objective (I can...): I can create my own loop in a program and choose what repeats and how many times. **CS concept:** Creating loops in a program · **Builds on:** Lesson 23 (repeat and count).

AI Connection (2 min, read aloud): The people who build AI set up its practice loops — what to repeat and how many rounds. *Ask the group:* "If you could set how many times a robot practices, what number would you pick?"

Materials: LEGO CS & AI kit + device per pair. **Model reuse:** Loops model from Lesson 21 — code only.

60-minute flow - Warm-up (10): "Your repeat" — each kid picks a move and a number ("jump, 3 times!"); the group does it. They're inventing loops. - **Concept + demo (8):** Show starting from an empty program: drop a repeat picture, choose the body, choose the count. Build one loop from scratch live. - **Build/Code (25):** Pairs create their own loop from scratch — they pick the actions for the body and the repeat number, then run and adjust. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Make a second, different loop in the same program and run them one after the other. - **Share + tidy-up (10):** Pairs show the loop they made and say what repeats and how many times. Model stays built.

Pair roles: Each partner creates one loop; swap at the wiggle break to try each other's. (*Trio: three loops, one each.*)

Differentiation: *Support* — choose the count for a loop body that's already there. *Extension / fast-finisher* — create two loops with different bodies. *Reteach* — build one simple loop together, then let them make the next alone.

Exit check: Pair creates a working loop from scratch and can say what repeats and how many times. **At-home (optional):** Invent a chore loop ("fold a shirt, repeat 5 times"); do it with a grown-up. ~5 min.
